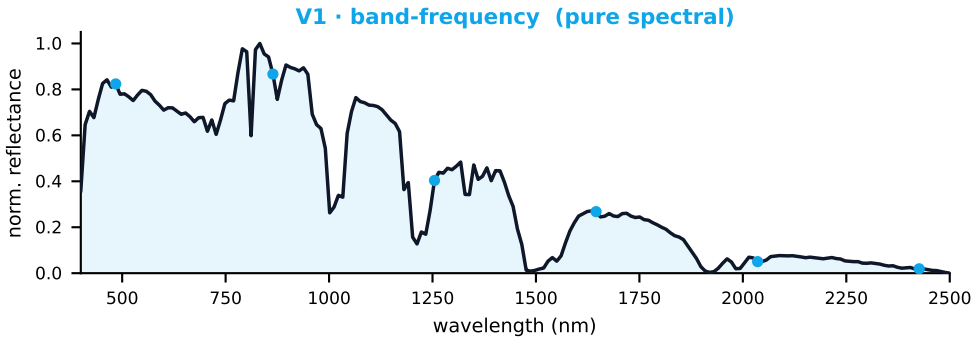


One spectrum, four token bags — the wordification step of LDA-on-HSI

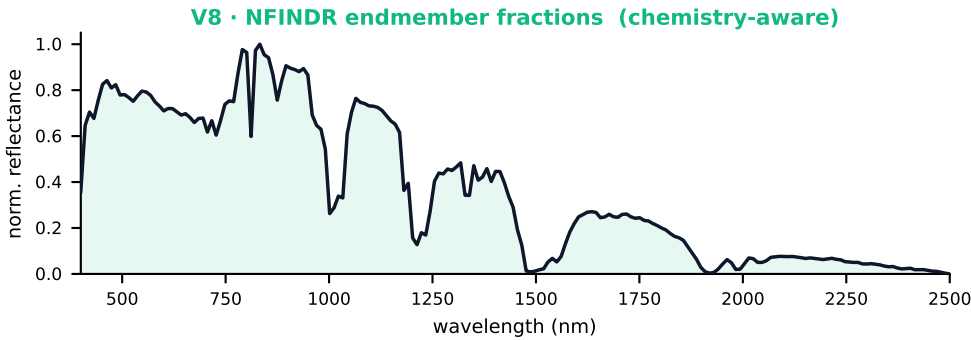
The identical pixel spectrum (left of each row) is turned into a bag-of-tokens by four recipes, one per design family. Source: real Soybean-notill pixel — Indian Pines (200 bands, 400-2500 nm).



token = band id · count = quantised intensity (0..7) · vocab = B = 200 bands



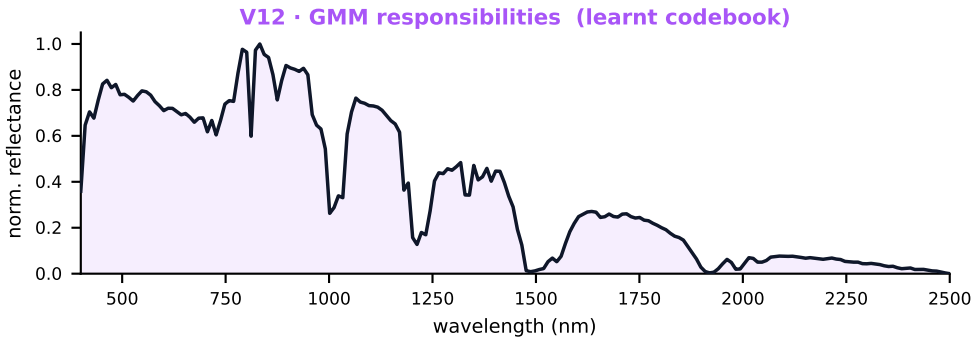
Every band fires once; its weight is the bin height. Canonical Procemin-2022 baseline — the reference all other recipes are scored against.



NFINDR extracts K=12 endmembers; NNLS unmix -> abundance fractions; each fraction quantised. token = endmember id
abundance simplex (sum=1)



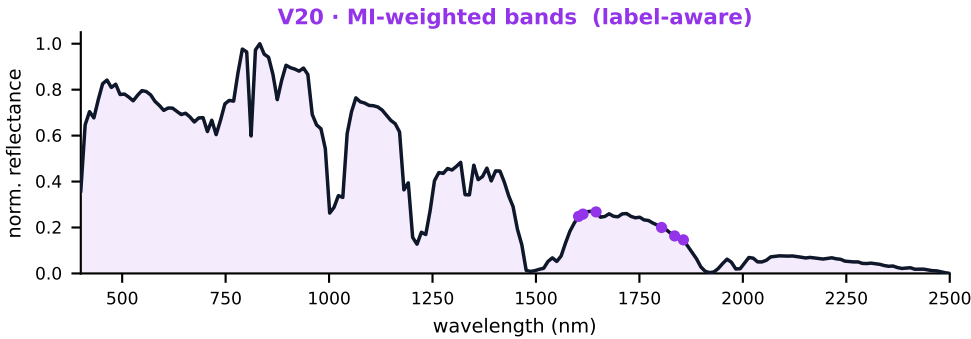
Only 6 of 12 endmembers carry weight in this pixel — the bag is short and chemistry-anchored.



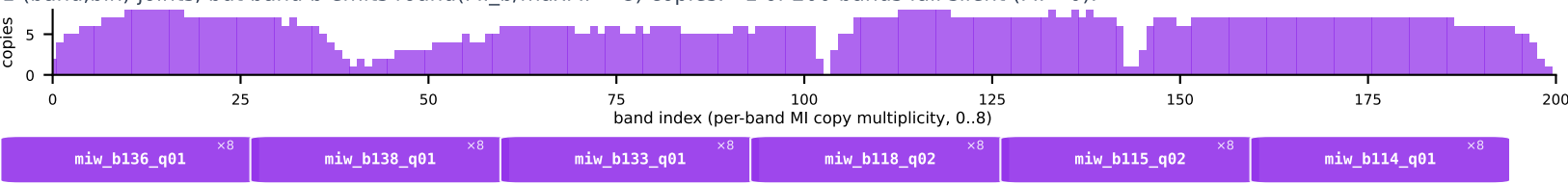
a 1-D GMM(Q=8) is fit on the intensities; each band's value is assigned a component g. token = (band, component) joint
bands per GMM component (B=200 total)



Bins are data-driven (EM-fit means) rather than equi-spaced — this is V12's only change versus V3's fixed alphabet.



V1 (band,bin) joints, but band b emits round(MI_b/maxMI × 8) copies. 1 of 200 bands fall silent (MI ≈0).



Discriminative bands dominate the bag; uninformative ones vanish. The only label-aware recipe in the sweep — P3's headline winner.